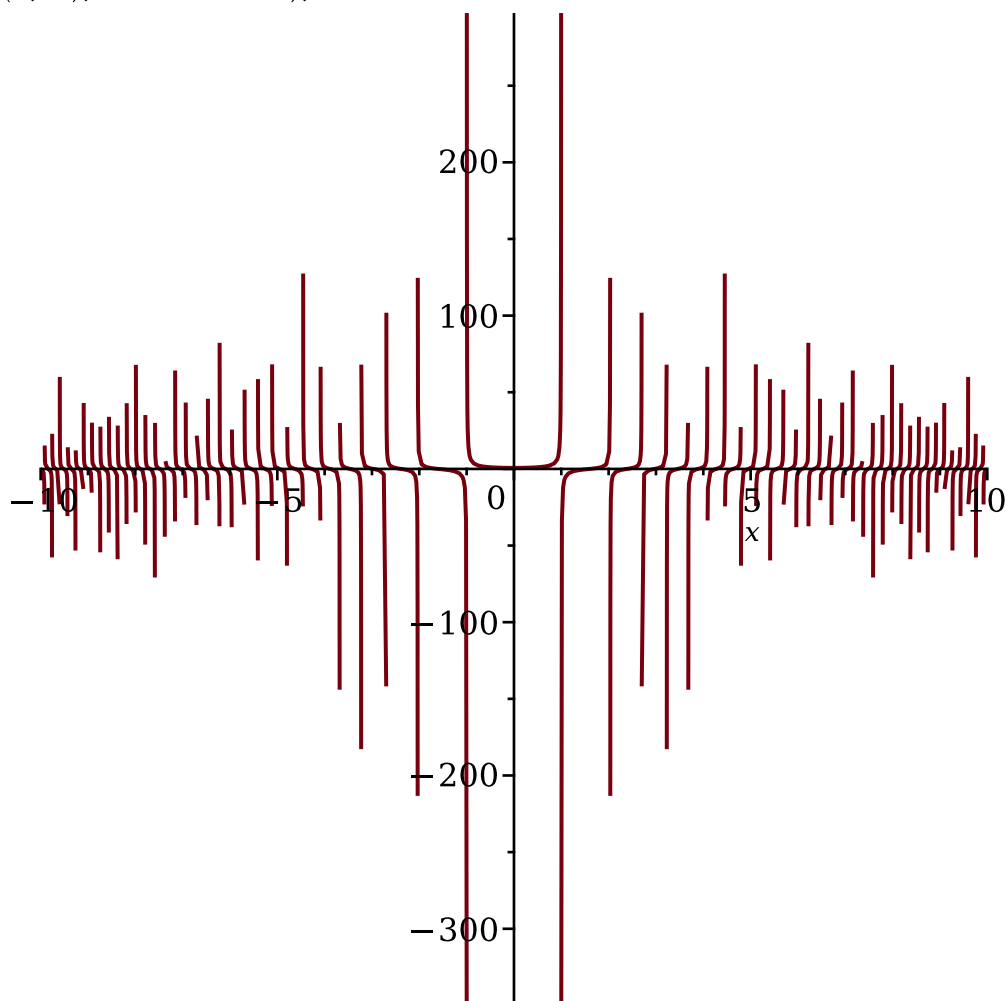


```

> with(DETools):
Error. (in dsolve) expecting an ODE or a set or list of ODEs.
Received ecdif1
> ecdif1 := diff(y(x), x) = 2·x·(1 + (y(x))2):
> soll := dsolve(ecdif1, y(x)):
> y1 := unapply(rhs(soll), x, c1)
      y1 := (x, c1) ↦ tan(x2 + 2·c1)
> plot(y1(x, 5), x = -10..10);

```

(1)

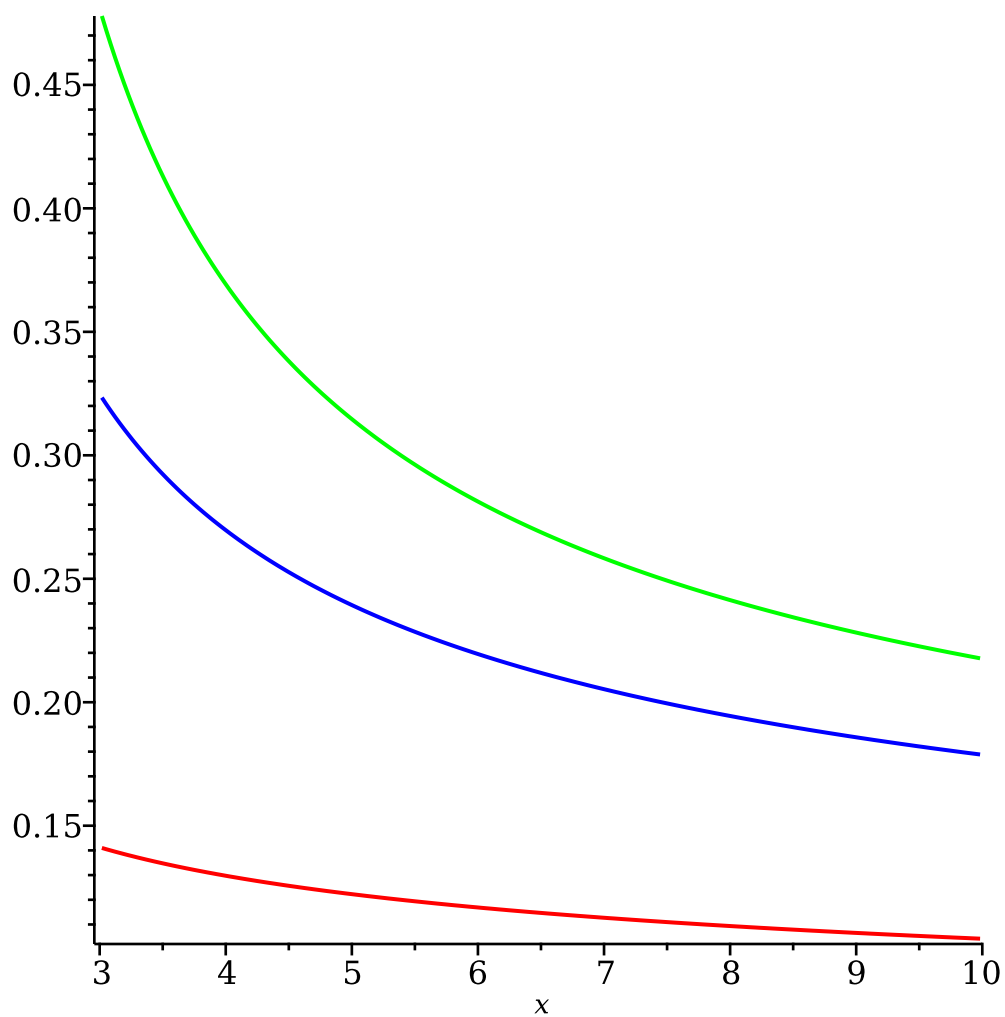


```

> restart:
> with(DETools):
> ecdif := (x2 - 1)·diff(y(x), x) + 2·x·(y(x))2 = 0:
> soll := dsolve(ecdif, y(x));
      soll := y(x) =  $\frac{1}{\ln(x-1) + \ln(x+1) + c_1}$ 
> y1 := unapply(rhs(soll), x, c1):
> plot([y1(x, 5), y1(x, 0), y1(x, 1)], x = 3..10, color = [red, green, blue]);

```

(2)



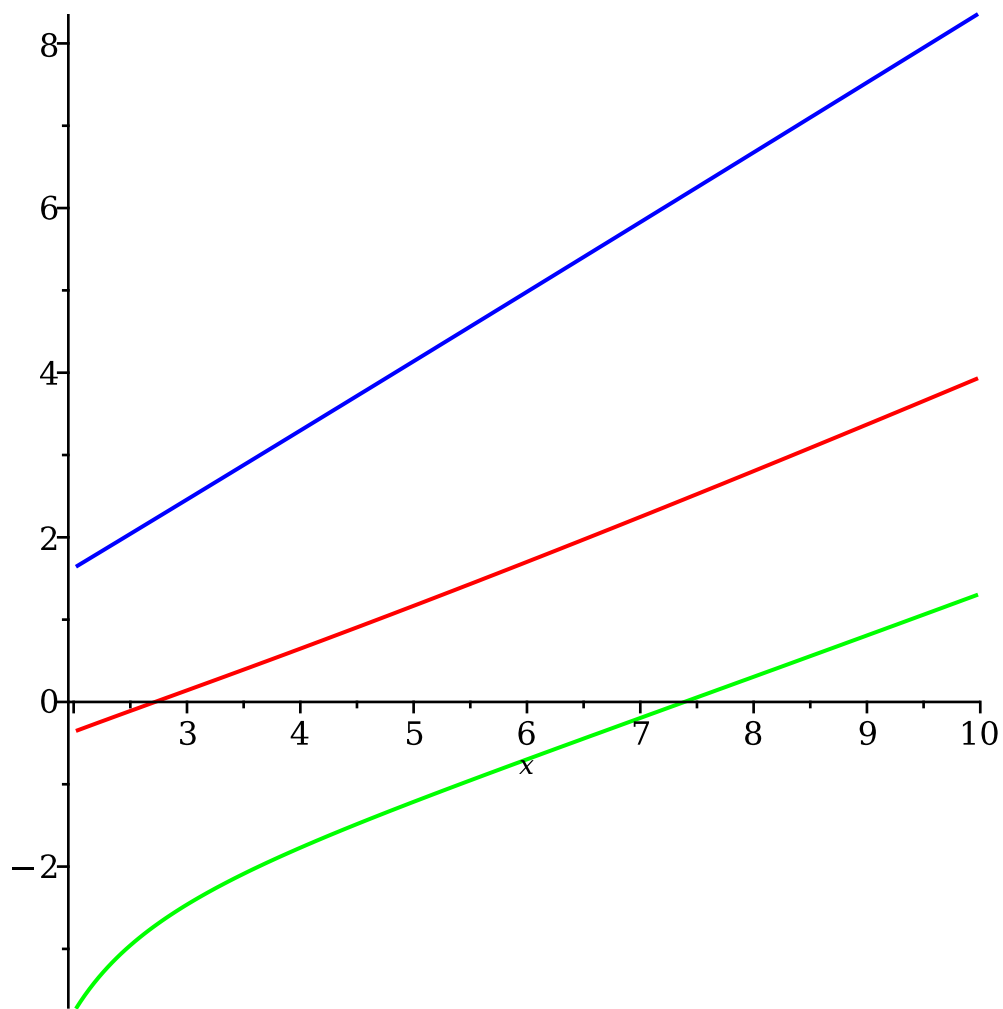
```

> restart:
> with(DETools):
> ecdif := 2*x^2*diff(y(x), x) = x^2 + (y(x))^2:
> soll := dsolve(ecdif, y(x));
      
$$soll := y(x) = \frac{x (\ln(x) + c_1 - 2)}{\ln(x) + c_1}$$

> y1 := unapply(rhs(soll), x, c_1):
> plot([y1(x, 1), y1(x, 0), y1(x, 10)], x = 2..10, color = [red, green, blue]);

```

(3)



```
> restart:
```

```
> with(DETools):
```

```
> ecdif := diff(y(x), x) = - x / y(x) :
```

```
> soll := dsolve(ecdif, y(x));
```

$$\text{soll} := y(x) = \sqrt{-x^2 + c_1}, y(x) = -\sqrt{-x^2 + c_1} \quad (4)$$

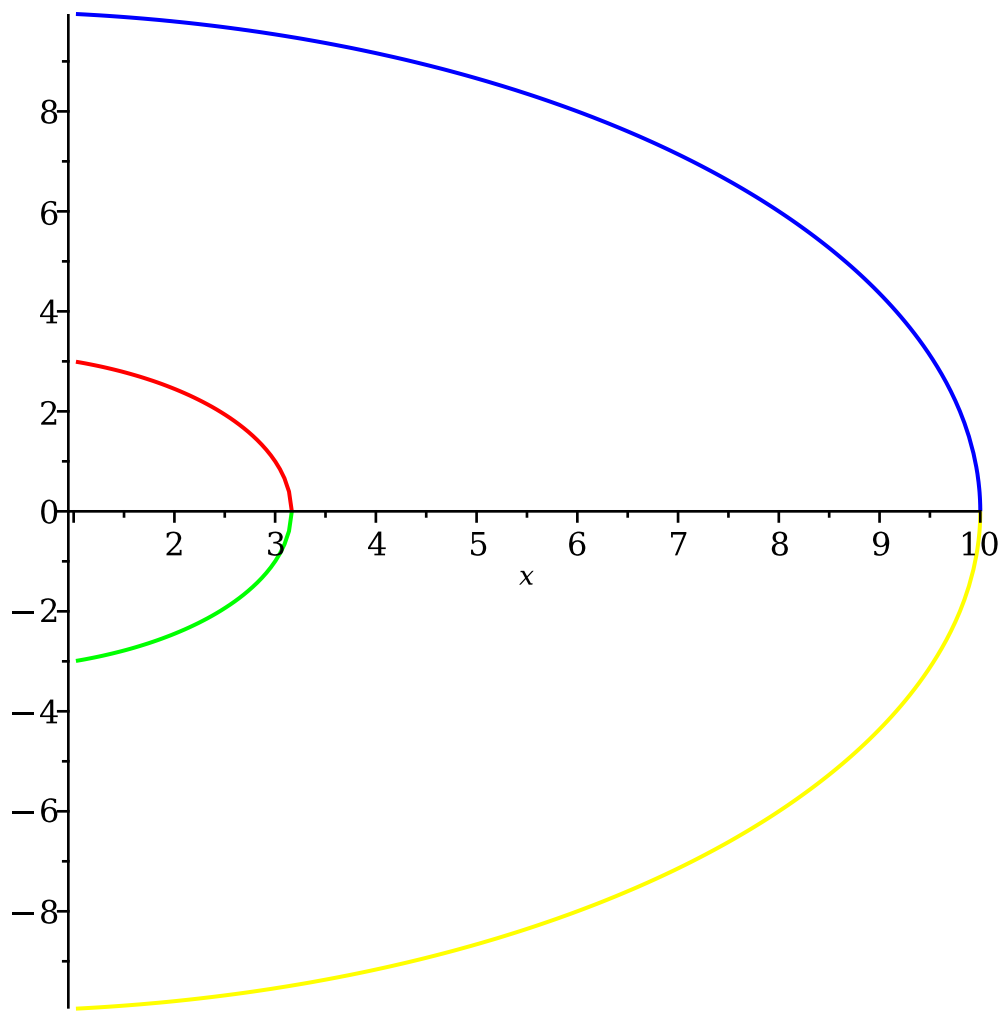
```
> y1 := unapply(rhs(soll[1]), x, c1);
```

$$y1 := (x, c_1) \mapsto \sqrt{-x^2 + c_1} \quad (5)$$

```
> y2 := unapply(rhs(soll[2]), x, c1);
```

$$y2 := (x, c_1) \mapsto -\sqrt{-x^2 + c_1} \quad (6)$$

```
> plot([y1(x, 10), y1(x, 100), y2(x, 10), y2(x, 100)], x = 1..10, color = [red, blue, green, yellow]);
```



```
> restart:
```

```
> with(DETools):
```

```
> ecdif := diff(y(x), x) = - x / (y(x))^3 :
```

```
> soll := dsolve(ecdif, y(x)) :
```

```
> y1 := unapply(rhs(soll[1]), x, c1);
```

$$y1 := (x, c_1) \mapsto (-2 \cdot x^2 + c_1)^{1/4} \quad (7)$$

```
> y2 := unapply(rhs(soll[2]), x, c1);
```

$$y2 := (x, c_1) \mapsto -(-2 \cdot x^2 + c_1)^{1/4} \quad (8)$$

```
> y3 := unapply(rhs(soll[3]), x, c1);
```

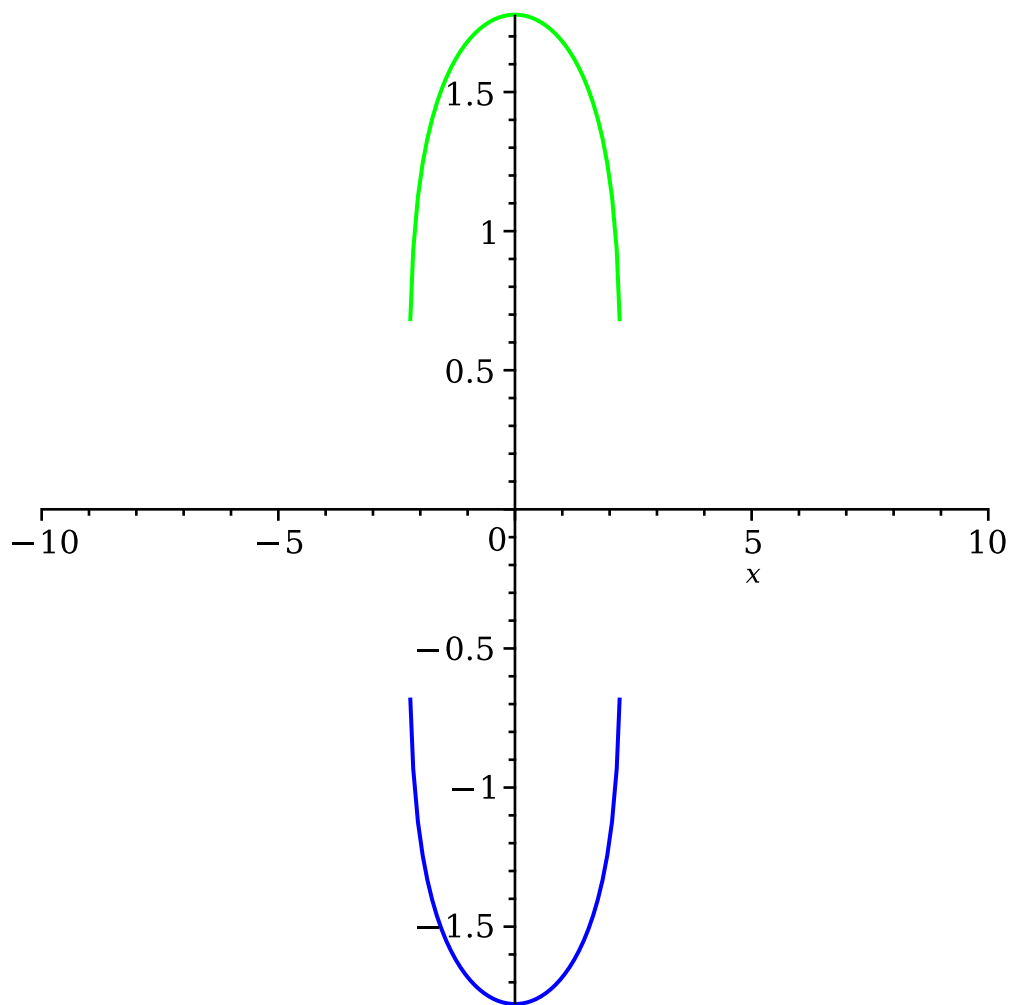
$$y3 := (x, c_1) \mapsto -I \cdot (-2 \cdot x^2 + c_1)^{1/4} \quad (9)$$

```
> y4 := unapply(rhs(soll[4]), x, c1);
```

$$y4 := (x, c_1) \mapsto I \cdot (-2 \cdot x^2 + c_1)^{1/4} \quad (10)$$

```
> plot([y1(x, 10), y2(x, 10), y3(x, 10), y4(x, 10)], x = -10..10, color = [green,
blue, red, yellow]);
```

Warning. unable to evaluate 2 of the 4 functions to numeric values
in the region; complex values were detected



```

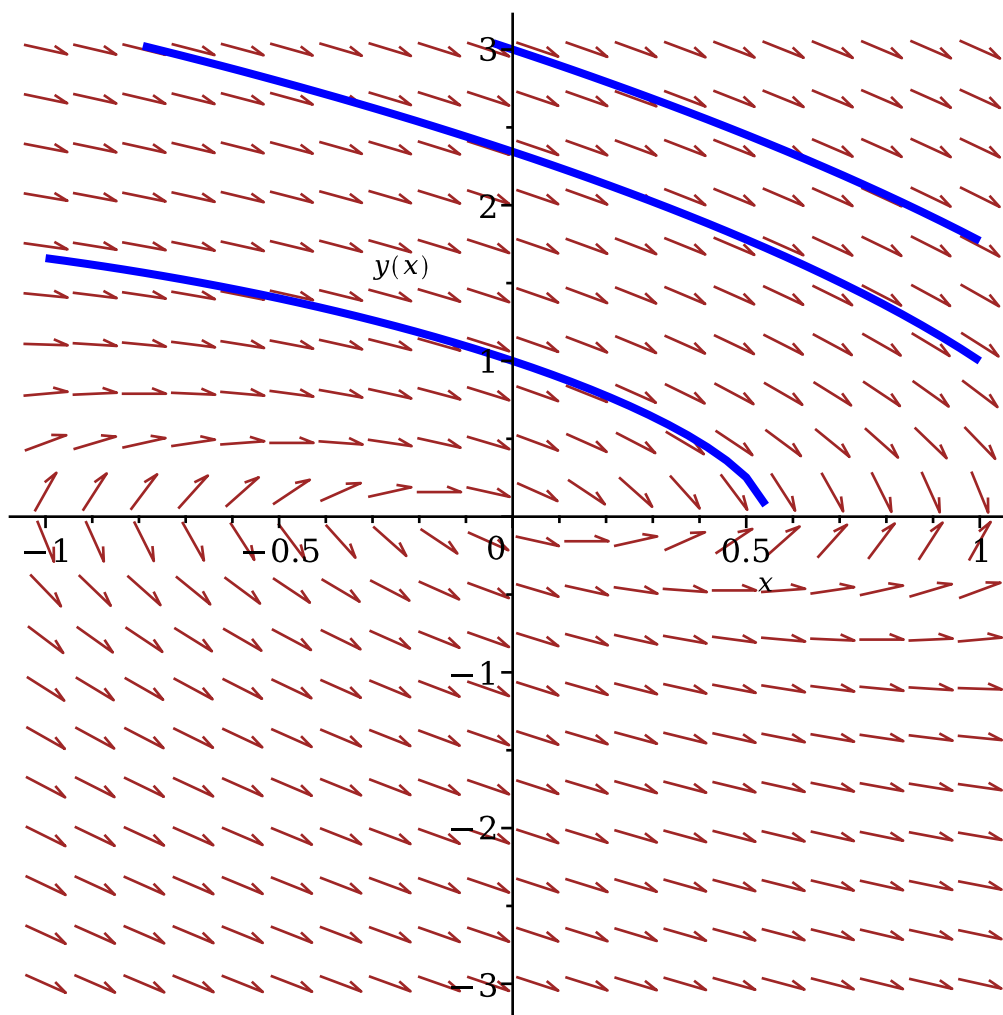
> restart:
> with(DETools):
> ecdif := y(x)·diff(y(x), x) = -x - y(x):
> soll := dsolve(ecdif, y(x), implicit);
soll := -  $\frac{\ln\left(\frac{x^2 + x y(x) + y(x)^2}{x^2}\right)}{2} + \frac{\sqrt{3} \arctan\left(\frac{(2 y(x) + x) \sqrt{3}}{3 x}\right)}{3} - \ln(x)$  (11)
- c1 = 0
>

```

```

>
> cond_in := [y(0) = 1], [y(1) = 1], [y(0) = 3]
cond_in := [y(0) = 1], [y(1) = 1], [y(0) = 3] (12)
> DEplot([ecdif], [y(x)], x = -1..1, y = -3..3, [cond_in], linecolor = blue);
Warning, plot may be incomplete, the following error(s) were issued:
cannot evaluate the solution further right of .54629311, probably
a singularity

```



```
> restart:
```

```
> with(DETools):
```

```
> ecdif := diff(y(x), x) + y(x)·tan(x) = 1/cos(x) :
```

```
> soll := dsolve(ecdif, y(x));
```

$$\text{soll} := y(x) = (\tan(x) + c_1) \cos(x)$$

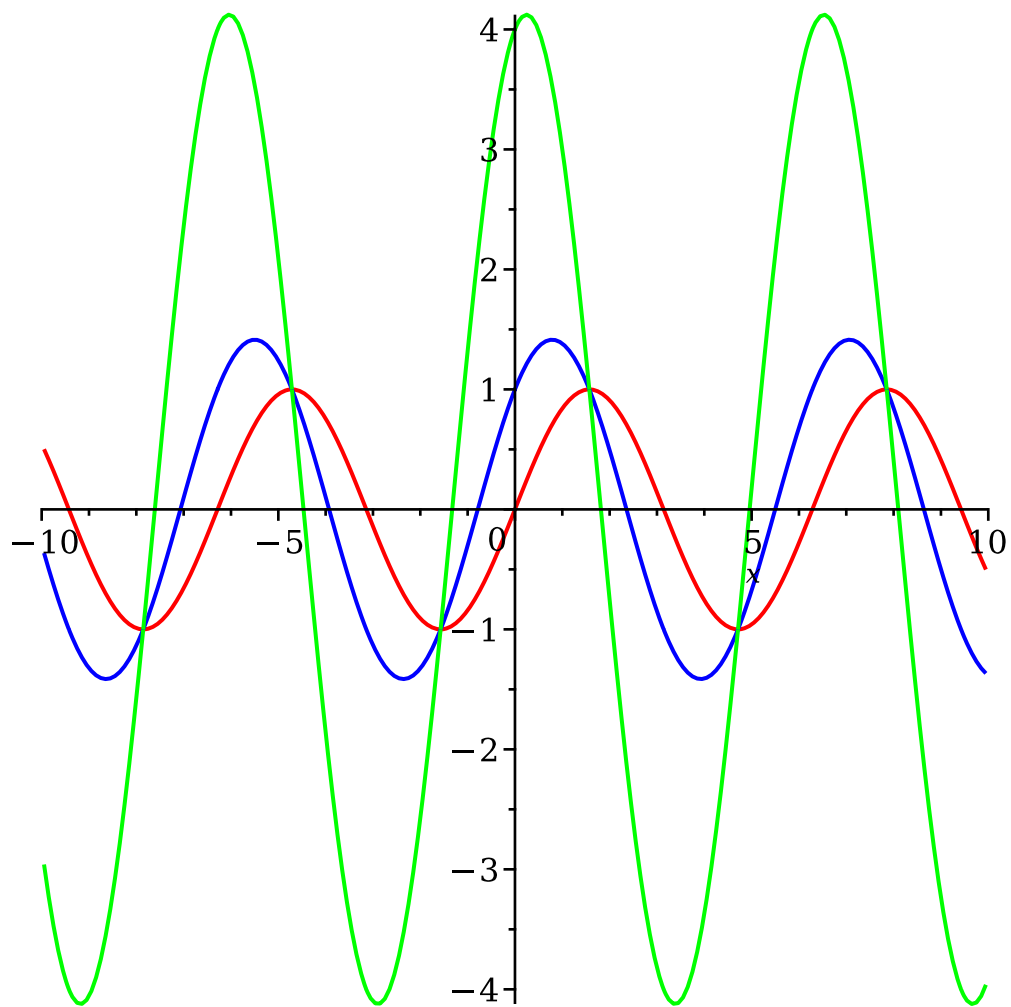
(13)

```
> y1 := unapply(rhs(soll), x, c1);
```

$$y1 := (x, c_1) \mapsto (\tan(x) + c_1) \cdot \cos(x)$$

(14)

```
> plot([y1(x, 0), y1(x, 1), y1(x, 4)], x = -10..10, color = [red, blue, green])
```



```
> restart:
```

```
> with(DETools):
```

```
> ecdf := diff(y(x), x) +  $\frac{2}{x} \cdot y(x) = x^3$ :
```

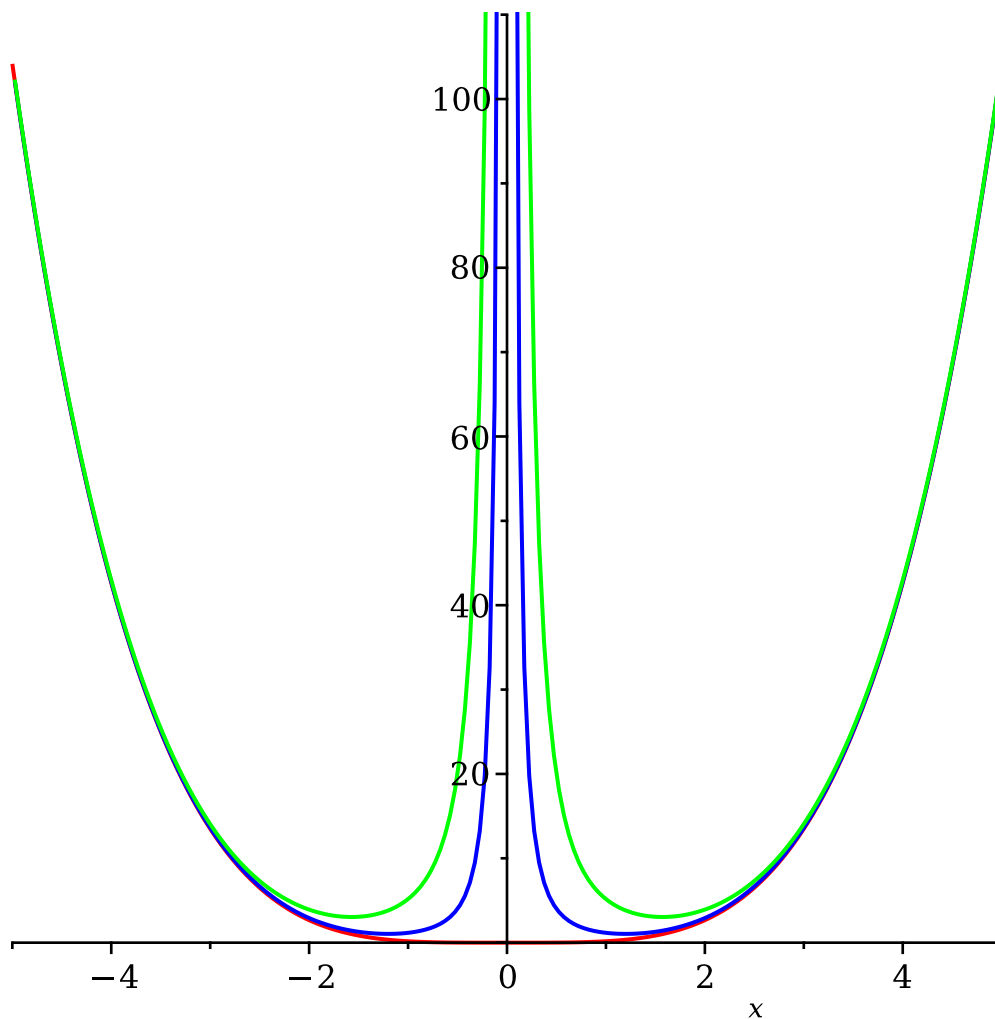
```
> soll := dsolve(ecdf, y(x));
```

$$soll := y(x) = \frac{\frac{x^6}{6} + c_1}{x^2}$$

(15)

```
> y1 := unapply(rhs(soll), x, c1):
```

```
> plot([y1(x, 0), y1(x, 1), y1(x, 5)], x = -5..5, color = [red, blue, green])
```

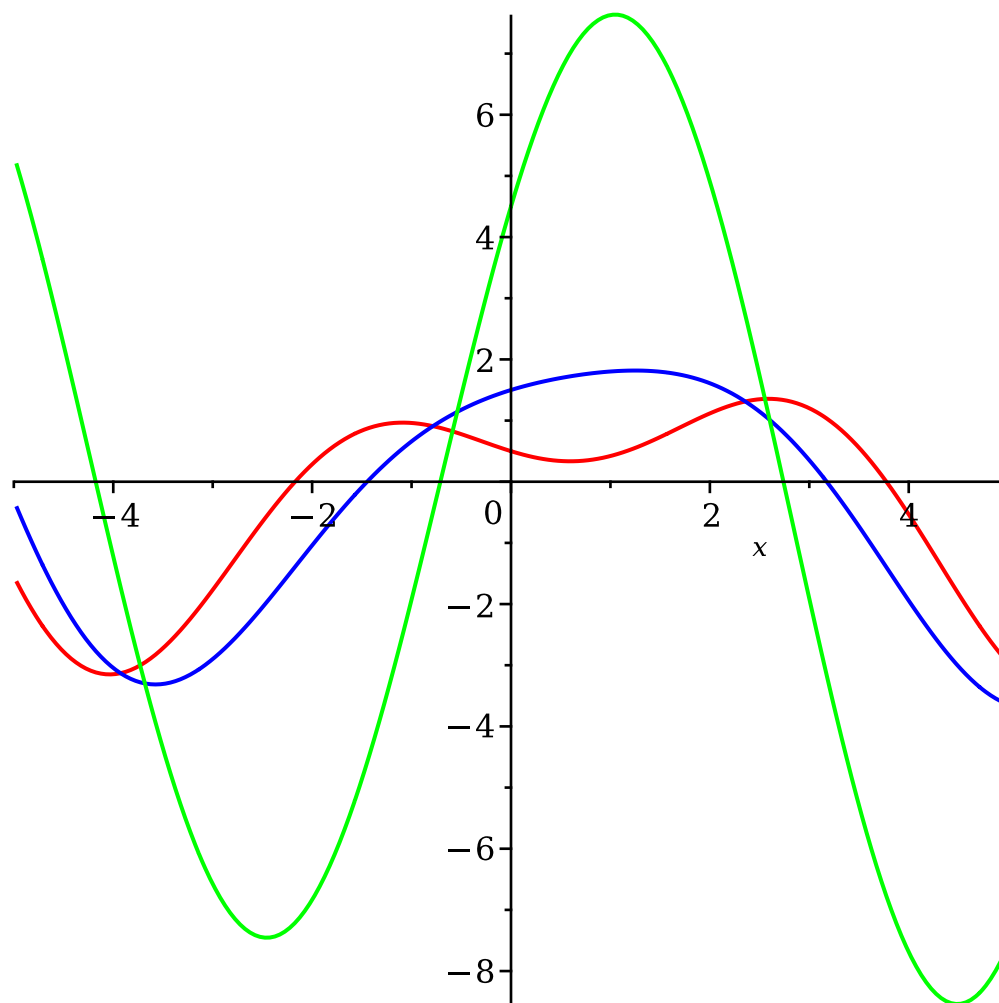


```

> with(DETools) :
> ecdif := diff(y(x), x, x) + y(x) = sin(x) + cos(x) :
> soll := dsolve(ecdif, y(x));
      
$$soll := y(x) = \sin(x) c_2 + \cos(x) c_1 + \frac{(1-x) \cos(x)}{2} + \frac{\sin(x) x}{2} \quad (16)$$

> y1 := unapply(rhs(soll), x, c1, c2) :
> plot([y1(x, 0, 0), y1(x, 1, 1), y1(x, 4, 6)], x = -5..5, color = [red, blue, green])

```

```

> restart:
> with(DETools):
> ecdif:=diff(y(x), x, x) - y(x) = exp(2·x):
> soll := dsolve(ecdif, y(x));

```

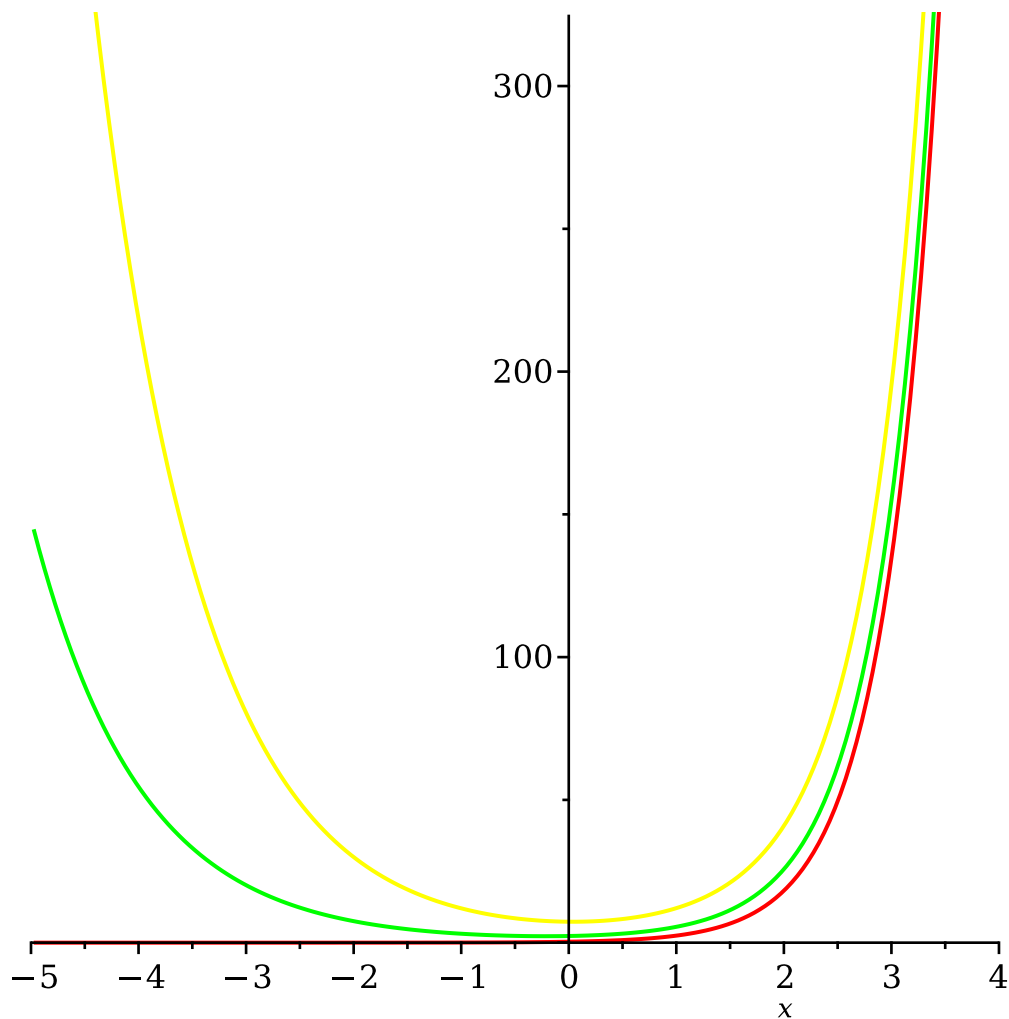
$$soll := y(x) = e^{-x} c_2 + e^x c_1 + \frac{e^{2x}}{3}$$

(17)

```

> y1 := unapply(rhs(soll), x, c1, c2):
> plot([y1(x, 0, 0), y1(x, 1, 1), y1(x, 3, 4)], x = -5..5, color = [red, green, yellow])

```



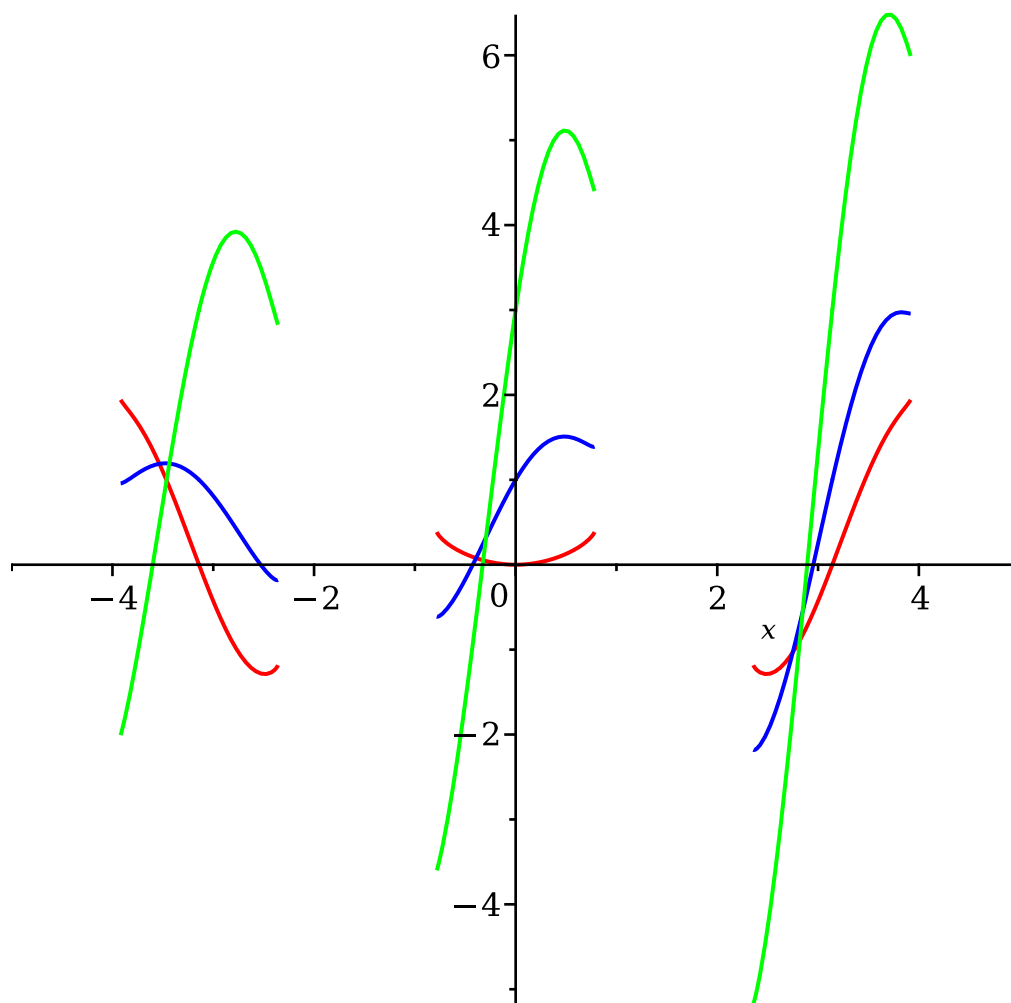
```

> restart:
> with(DETools):
  
$$ecdif := \text{diff}(y(x), x, x) + 4 \cdot y(x) = \frac{1}{\cos(2 \cdot x)} :$$

> soll := dsolve(ecdif, y(x));
  
$$soll := y(x) = \sin(2 x) c_2 + \cos(2 x) c_1 + \frac{x \sin(2 x)}{2} + \frac{\ln(\cos(2 x)) \cos(2 x)}{4} \quad (18)$$

> y1 := unapply(rhs(soll), x, c1, c2):
> plot([y1(x, 0, 0), y1(x, 1, 1), y1(x, 3, 4)], x = -5..5, color = [red, blue, green])

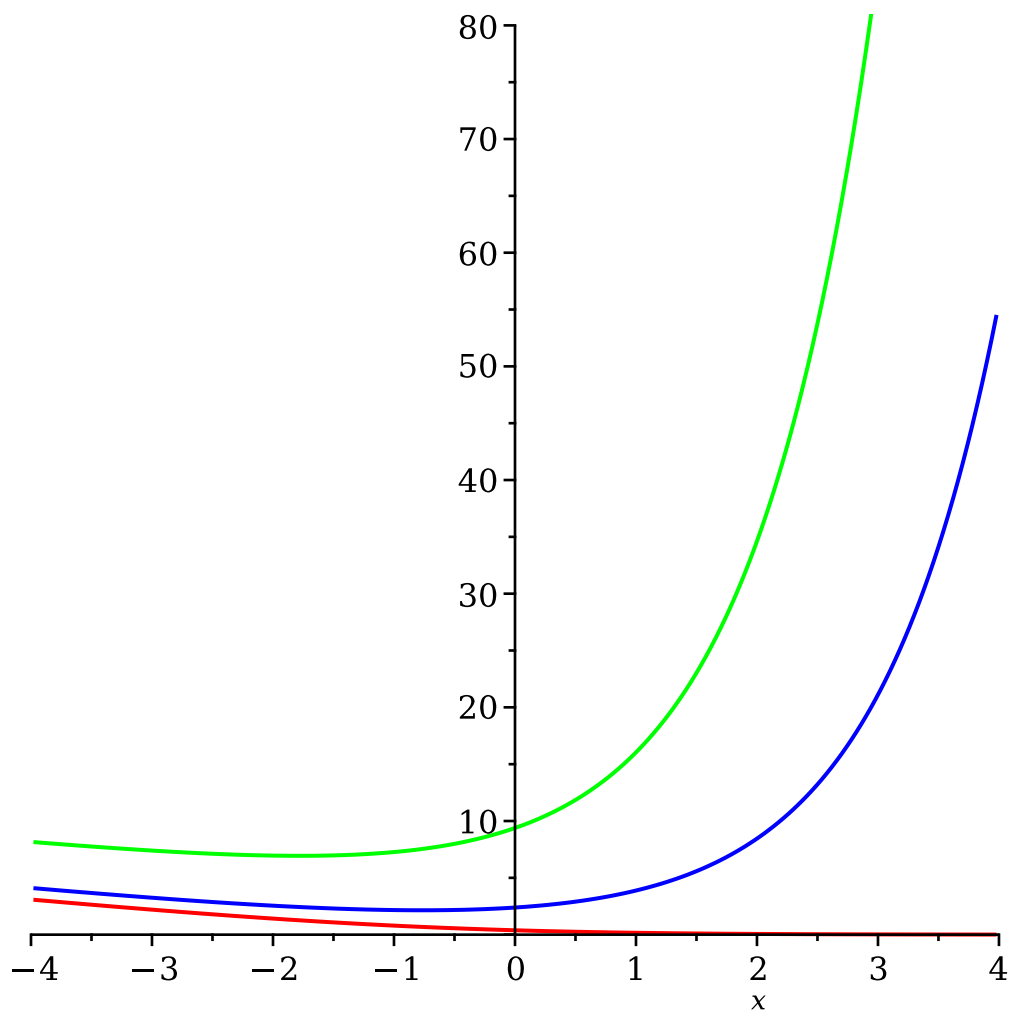
```



```

> restart:
> with(DETools):
> ecdif := diff(y(x), x, x) - diff(y(x), x) = 1 / (1 + exp(x)) :
> soll := dsolve(ecdif, y(x))
      soll := y(x) = -x + e^x c_1 + ln(1 + e^x) (1 + e^x) - 1 - e^x ln(e^x) + c_2      (19)
> y1 := unapply(rhs(soll), x, c_1, c_2):
> plot([y1(x, 0, 0), y1(x, 1, 1), y1(x, 4, 5)], x = -4..4, color = [red, blue, green])

```



```

> restart:
> with(DETools):
> ecdif := diff(y(x), x) = 1 + (y(x))^2:
> cond_in := y(0) = 1:
> soll := dsolve({ecdif, cond_in}, y(x));

```

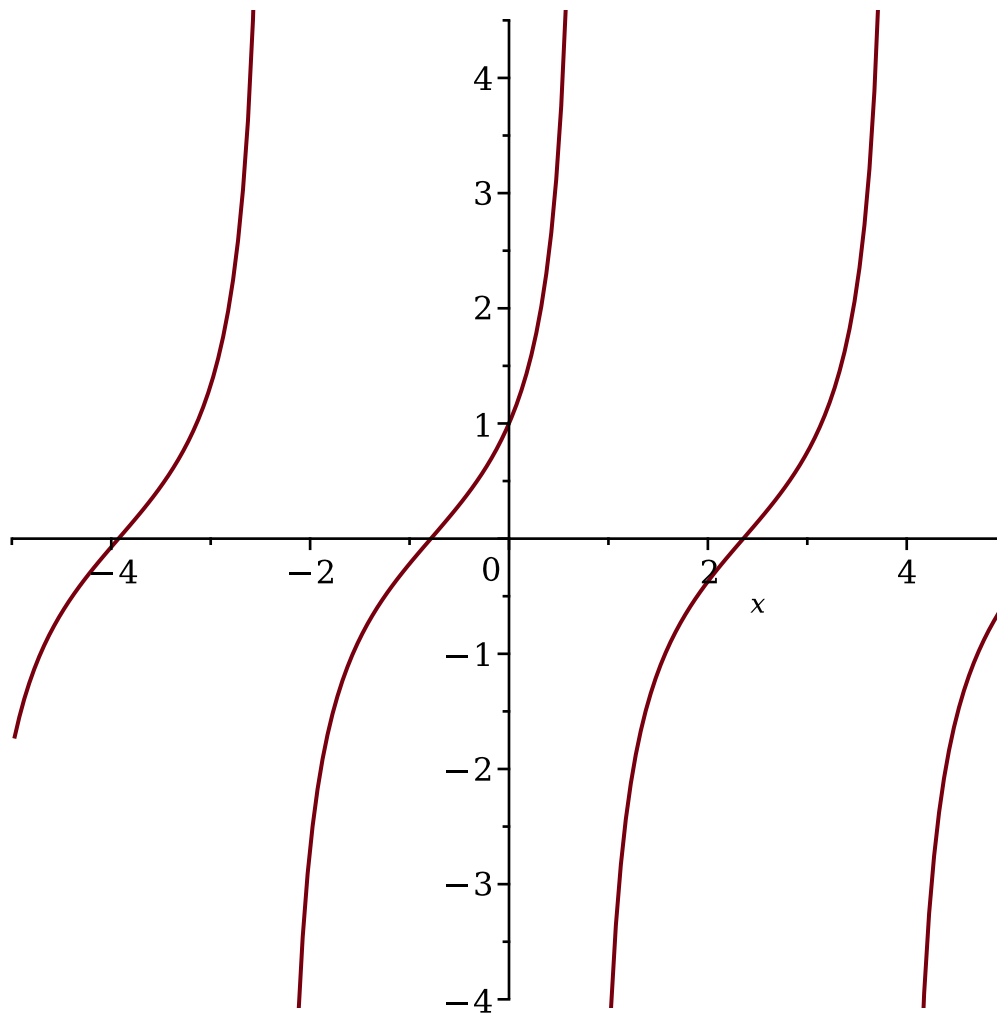
$$soll := y(x) = \tan\left(x + \frac{\pi}{4}\right)$$

(20)

```

> y1 := unapply(rhs(soll), x):
> plot(y1(x), x = -5..5)

```



```
> restart:
```

```
> with(DETools):
```

```
> ecdif := diff(y(x), x) =  $\frac{1}{1-x^2} \cdot y(x) + 1 + x$ :
```

```
> cond_in := y(0) = 0:
```

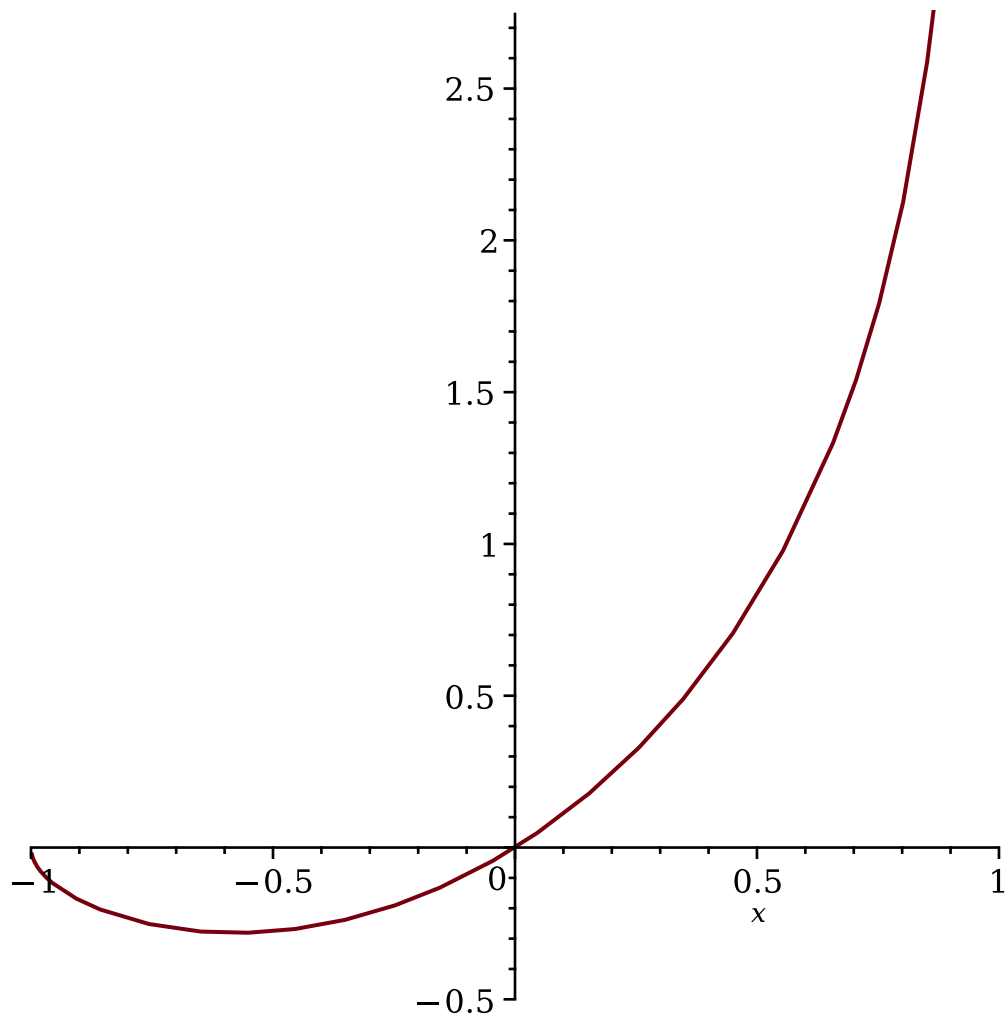
```
> soll := dsolve({ecdif, cond_in}, y(x));
```

$$soll := y(x) = \frac{(x\sqrt{-x^2+1} + \arcsin(x))(x+1)}{2\sqrt{-x^2+1}}$$

(21)

```
> y1 := unapply(rhs(soll), x):
```

```
> plot(y1(x), x = -10..10)
```

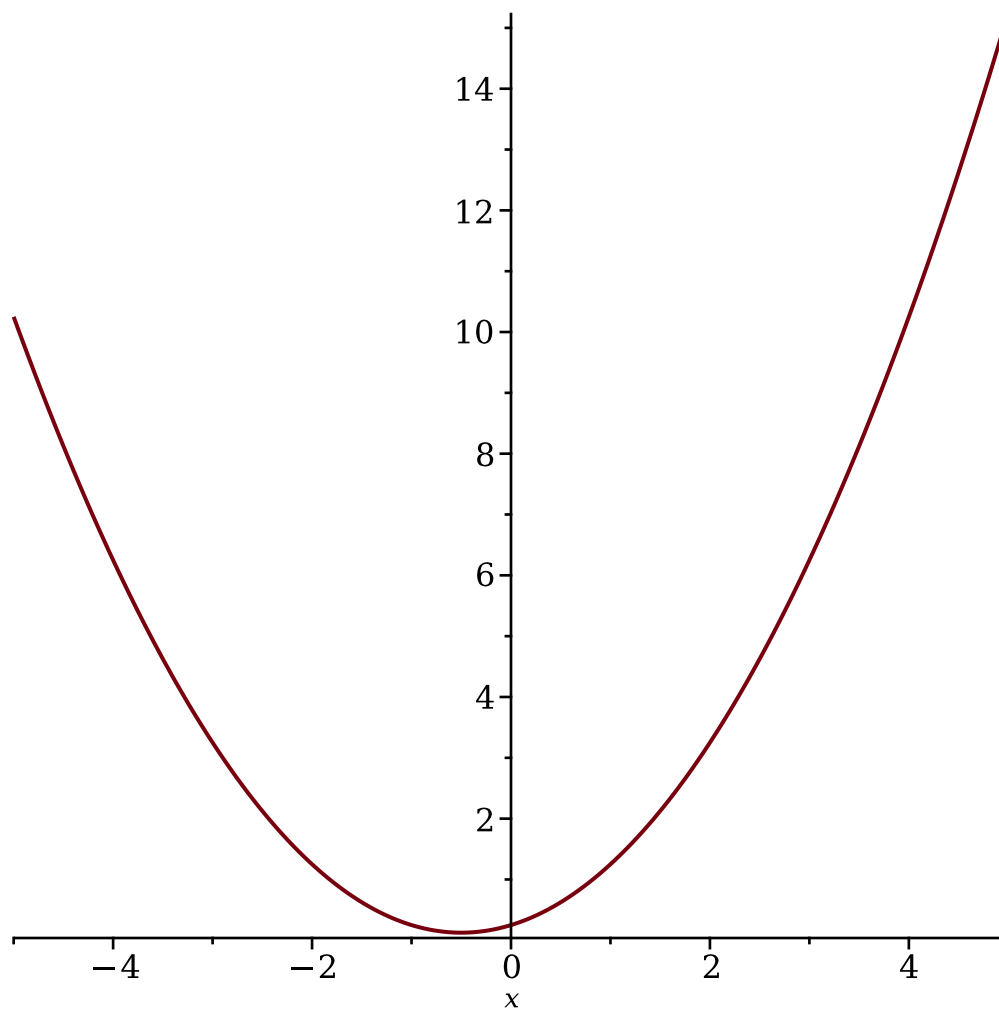


```

> restart:
> with(DETools):
> ecdf := diff(y(x), x) - 2·y(x) = -x2:
> cond_in := y(0) = 1/4:
> soll := dsolve({ecdf, cond_in}, y(x));
               soll := y(x) = 1/2 x2 + 1/2 x + 1/4
> y1 := unapply(rhs(soll), x):
  plot(y1(x), x = -5..5);

```

(22)

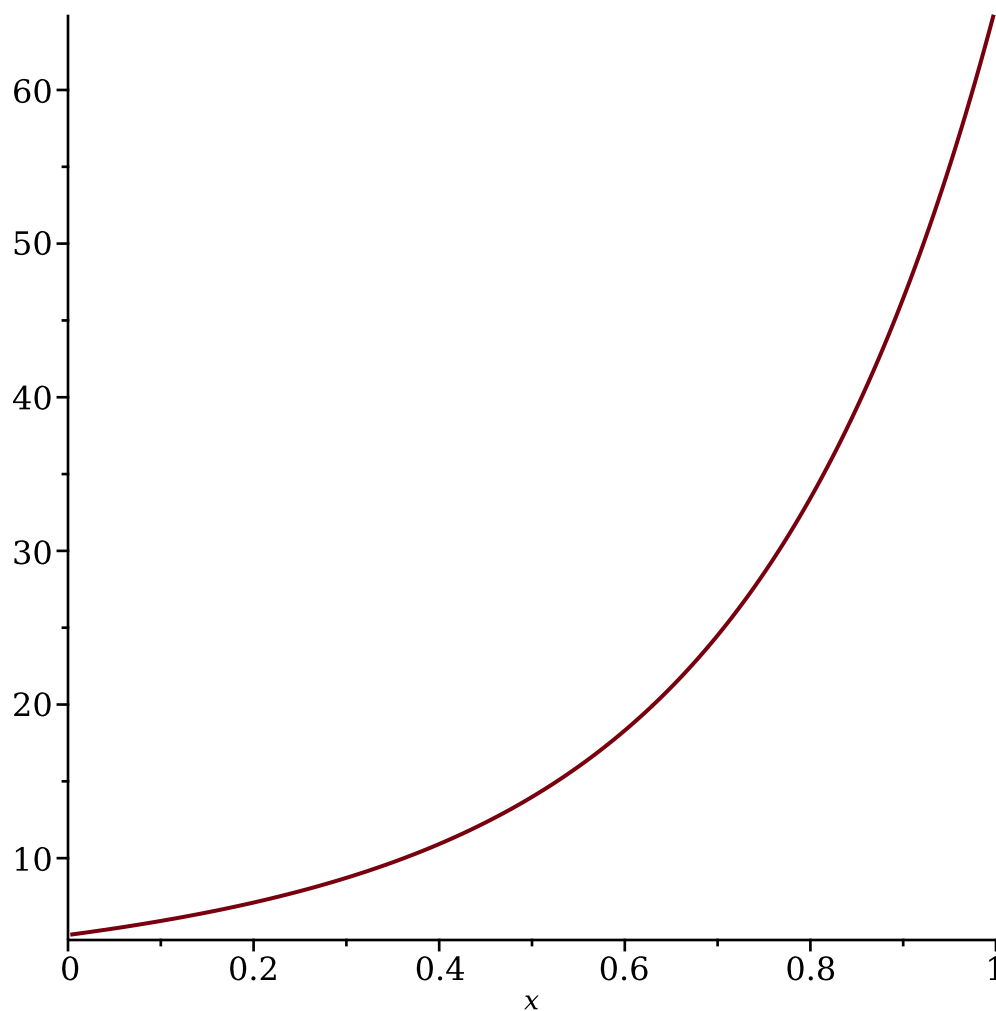


```

> restart:
> with(DETools):
> ecdif := diff(y(x), x, x) - 5·diff(y(x), x) + 4·y(x) = 0:
> cond_in := y(0) = 5, D(y)(0) = 8:
> soll := dsolve({ecdif, cond_in}, y(x));
                $soll := y(x) = e^{4x} + 4e^x$ 
> y1 := unapply(rhs(soll), x):
> plot(y1(x), x = 0..1)

```

(23)

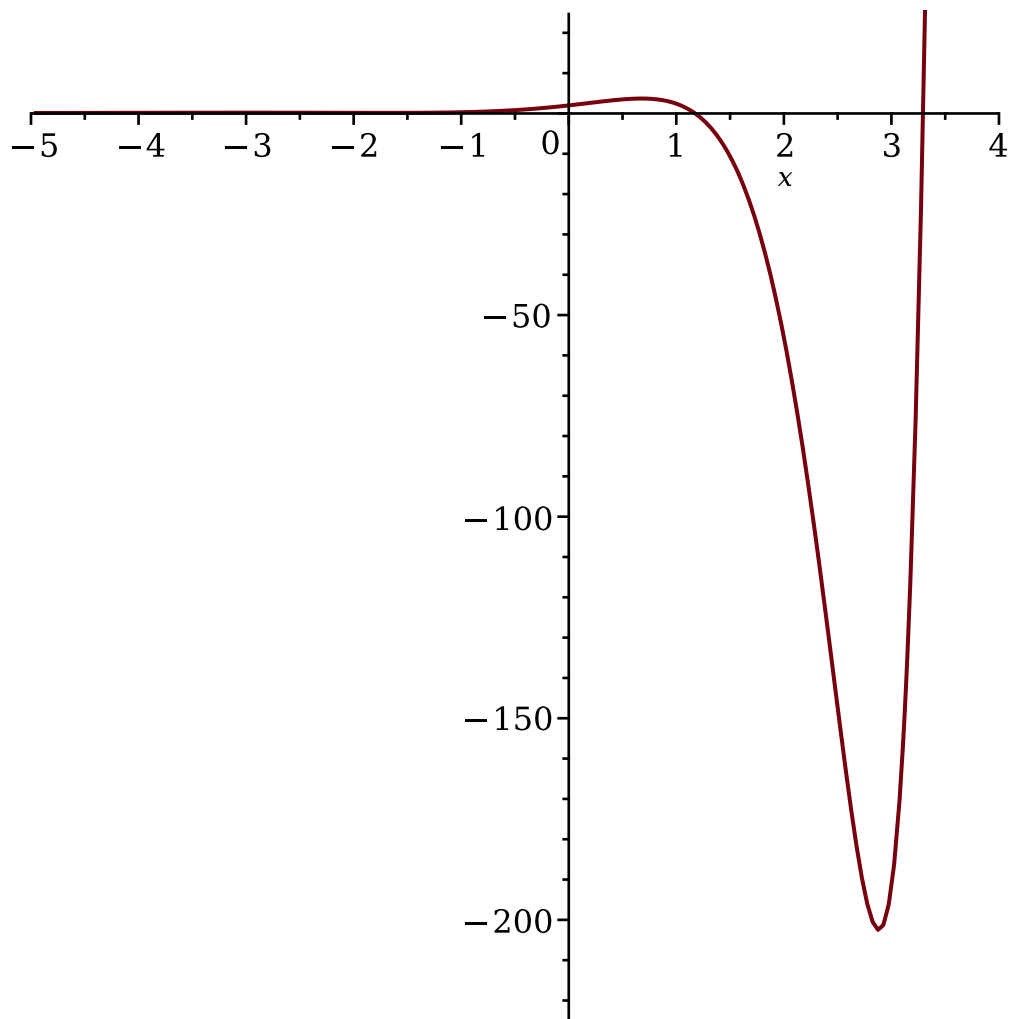


```

> restart:
> with(DETools):
> ecdif := diff(y(x), x, x) - 4·diff(y(x), x) + 5·y(x) = 2·x2·exp(x):
> cond_in := y(0) = 2, D(y)(0) = 3:
> soll := dsolve({ecdif, cond_in}, y(x));
            $soll := y(x) = (-2 \sin(x) + \cos(x)) e^{2x} + (x + 1)^2 e^x$ 
> y1 := unapply(rhs(soll), x):
> plot(y1(x), x = -5..5);

```

(24)



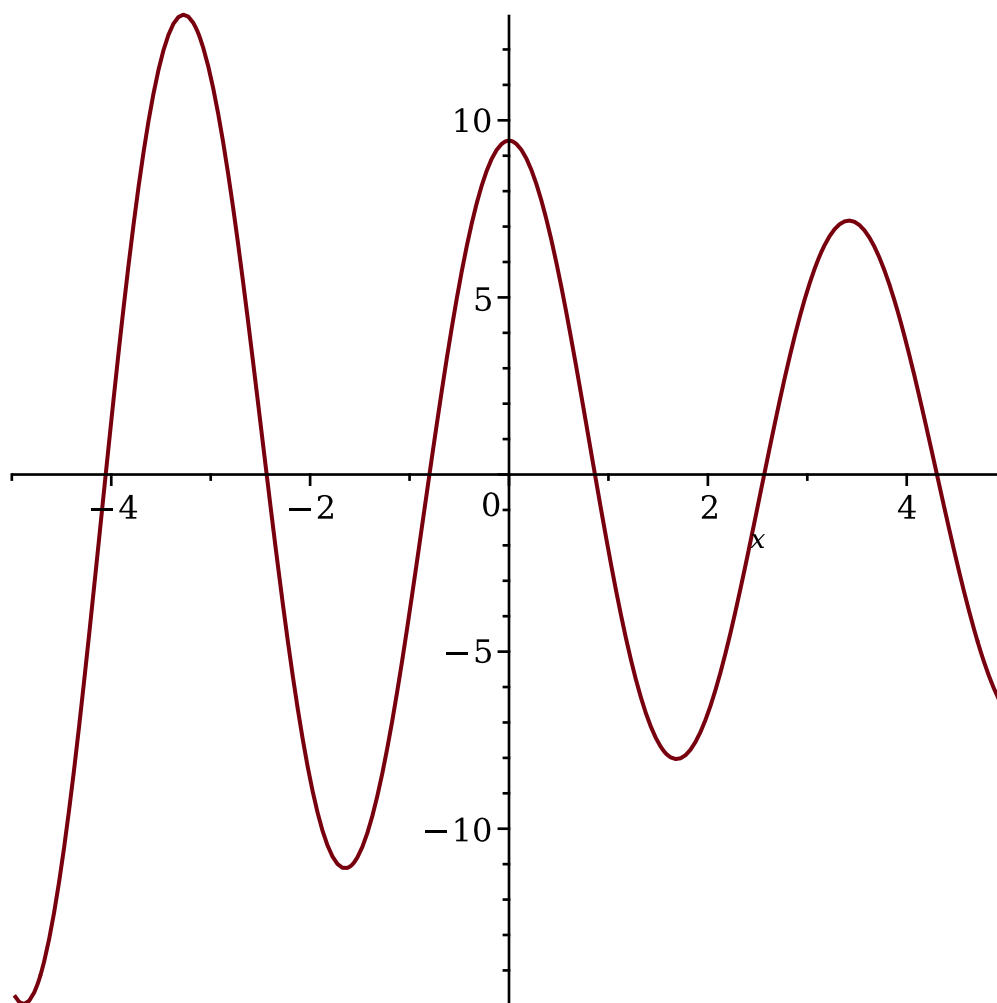
```

> restart:
> with(DETools):
> ecdif:=diff(y(x), x, x) + 4*y(x) = 4*(sin(2*x) + cos(2*x)):
> cond_in := y(Pi) = 2*Pi, D(y)(Pi) = 2*Pi:
> soll := dsolve({ecdif, cond_in}, y(x));
      
$$soll := y(x) = (-x + 3\pi) \cos(2x) + \frac{\sin(2x)(2x + 1)}{2}$$

> y1 := unapply(rhs(soll), x):
> plot(y1(x), x=-5..5);

```

(25)



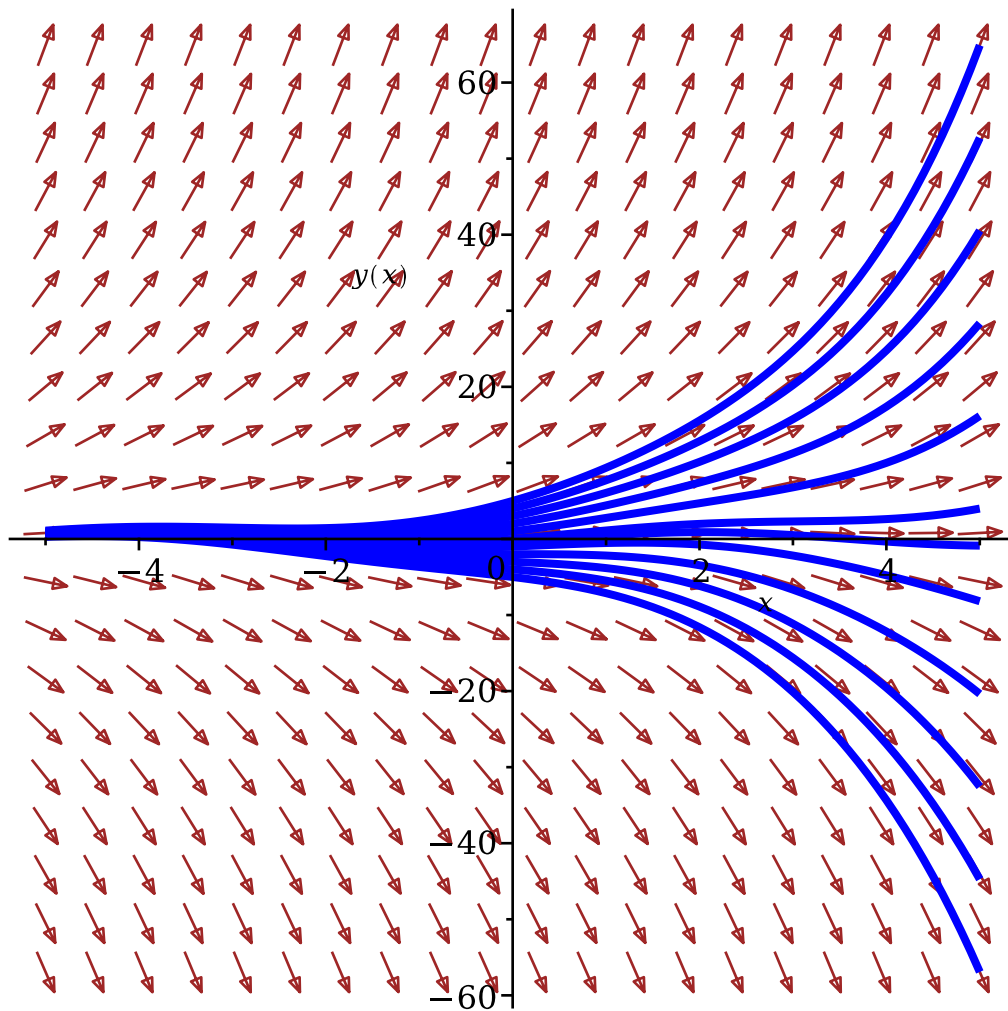
```
> restart:
```

```
> with(DETools):
```

```
> ecdif := diff(y(x), x) -  $\frac{1}{2}$  · y(x) = cos(x) :
```

```
> cond_in := y(0) = 1, y(0) = 2, y(0) = 3, y(0) = 4, y(0) = 5, y(0) = -1, y(0) = -2,
y(0) = -3, y(0) = -4, y(0) = -5, y(0) = 0, y(0) =  $-\frac{2}{5}$  :
```

```
> DEplot([ecdif], [y(x)], x = -5..5, [cond_in], arrows = medium, linecolor
= blue, stepsize = 0.1)
```



```
> soll := dsolve(ecdif, y(x));
```

$$\text{soll} := y(x) = -\frac{2 \cos(x)}{5} + \frac{4 \sin(x)}{5} + e^{\frac{x}{2}} c_1 \quad (26)$$

```
> a_0 := -\frac{2}{5};
```

$$a_0 := -\frac{2}{5} \quad (27)$$

```
> # y(0) = a <=> c_1 - \frac{2}{5} = a \rightarrow c = a + \frac{2}{5} \rightarrow c = 0 \text{ when } a = -\frac{2}{5}
```

```
> restart:
```

```
> with(DETools):
```

```
> ecdif := diff(y(x), x) + \frac{k}{x} \cdot y(x) = x^3:
```

```
> soll := dsolve(ecdif, y(x));
```

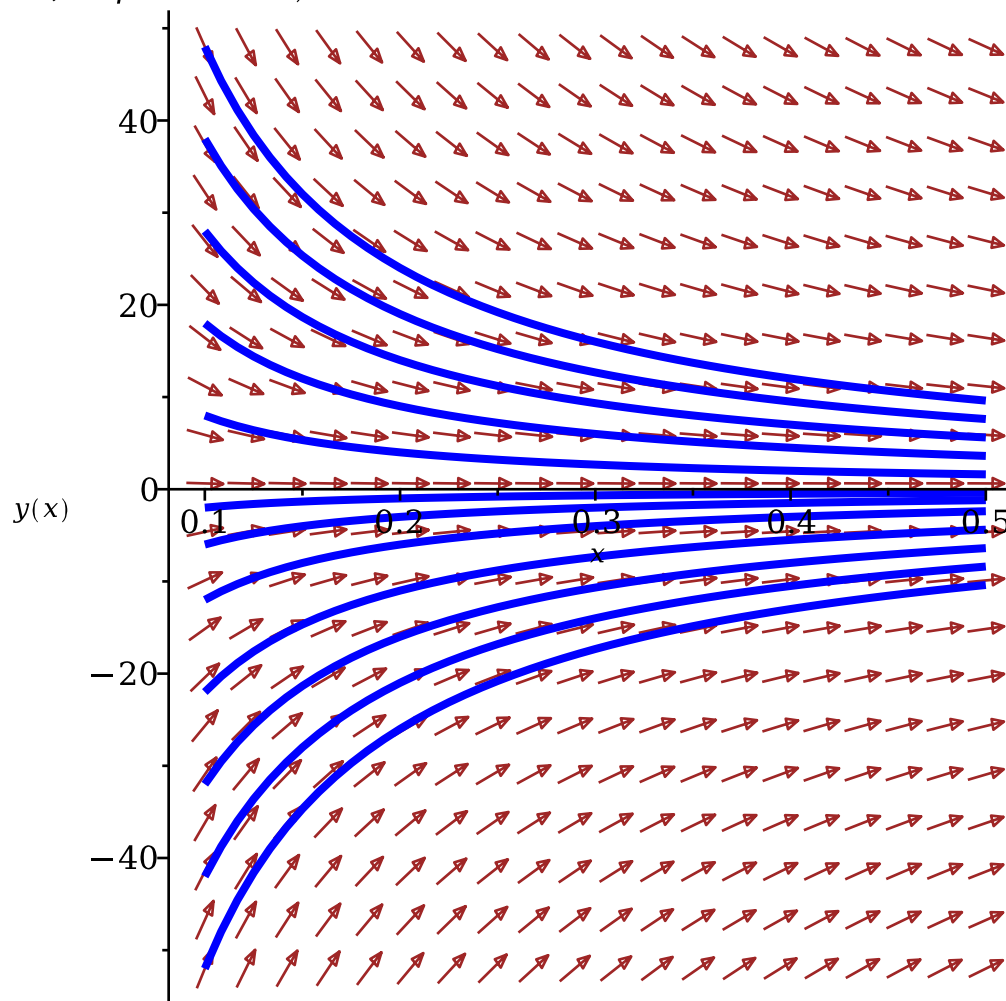
$$\text{soll} := y(x) = \frac{x^4}{4+k} + x^{-k} c_1 \quad (28)$$

```
> ecdif2 := diff(y(x), x) +  $\frac{1}{x}$  · y(x) = x3 :
```

Error. (in DEtools/dfieldplot) invalid use of initial points or option - see phaseportrait

```
> cond_in := y(1) = 1, y(1) = 2, y(1) = 3, y(1) = 4, y(1) = 5, y(1) = -1, y(1) = -2,
y(1) = -3, y(1) = -4, y(1) = -5, y(1) = 0, y(1) =  $-\frac{2}{5}$  :
```

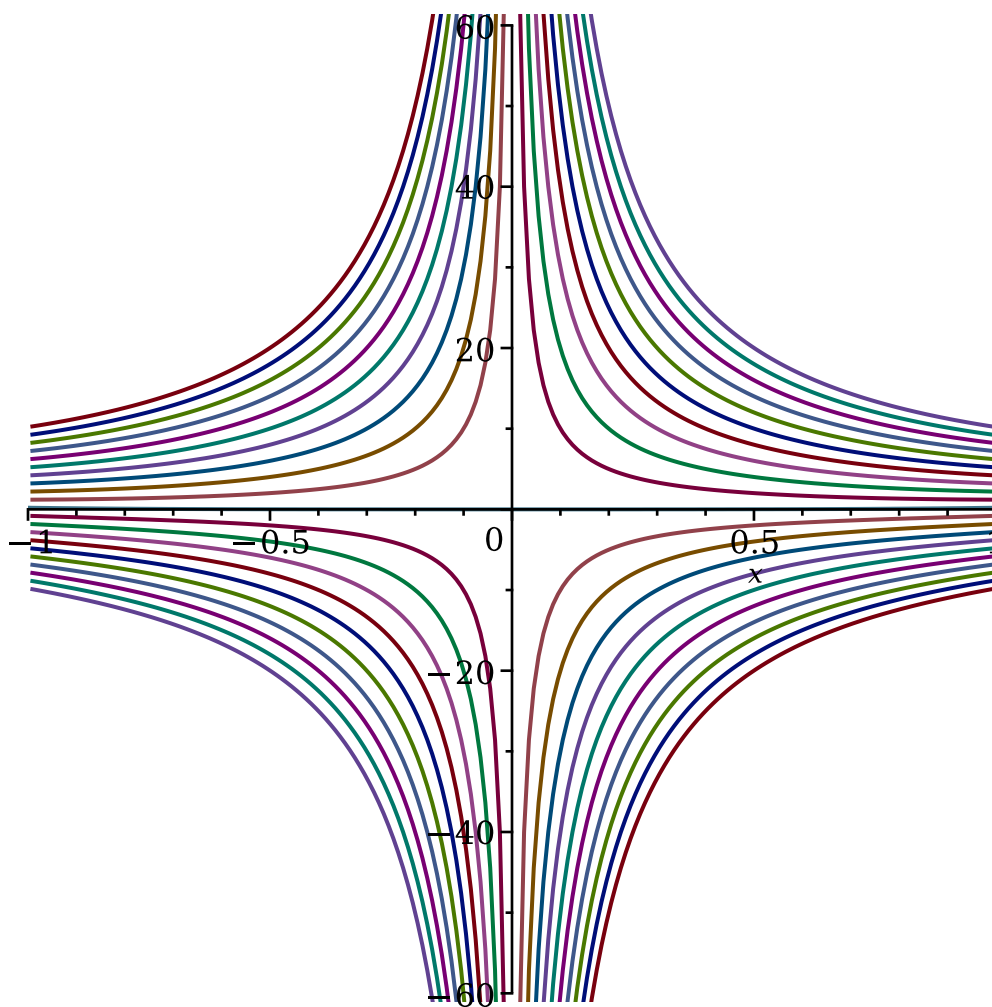
```
DEplot([ecdif2], [y(x)], x = 0.1 .. 0.5, [cond_in], arrows = medium, linecolor
= blue, stepsize = 0.1)
```



```
> y1 := unapply(rhs(soll), x, k, c1) :
```

```
> list_f := y1(x, 1, i) $ i = -10 .. 10 :
```

```
> plot([list_f], x = -1 .. 1)
```



```
> cond_in2 := y(1) = 0:
```

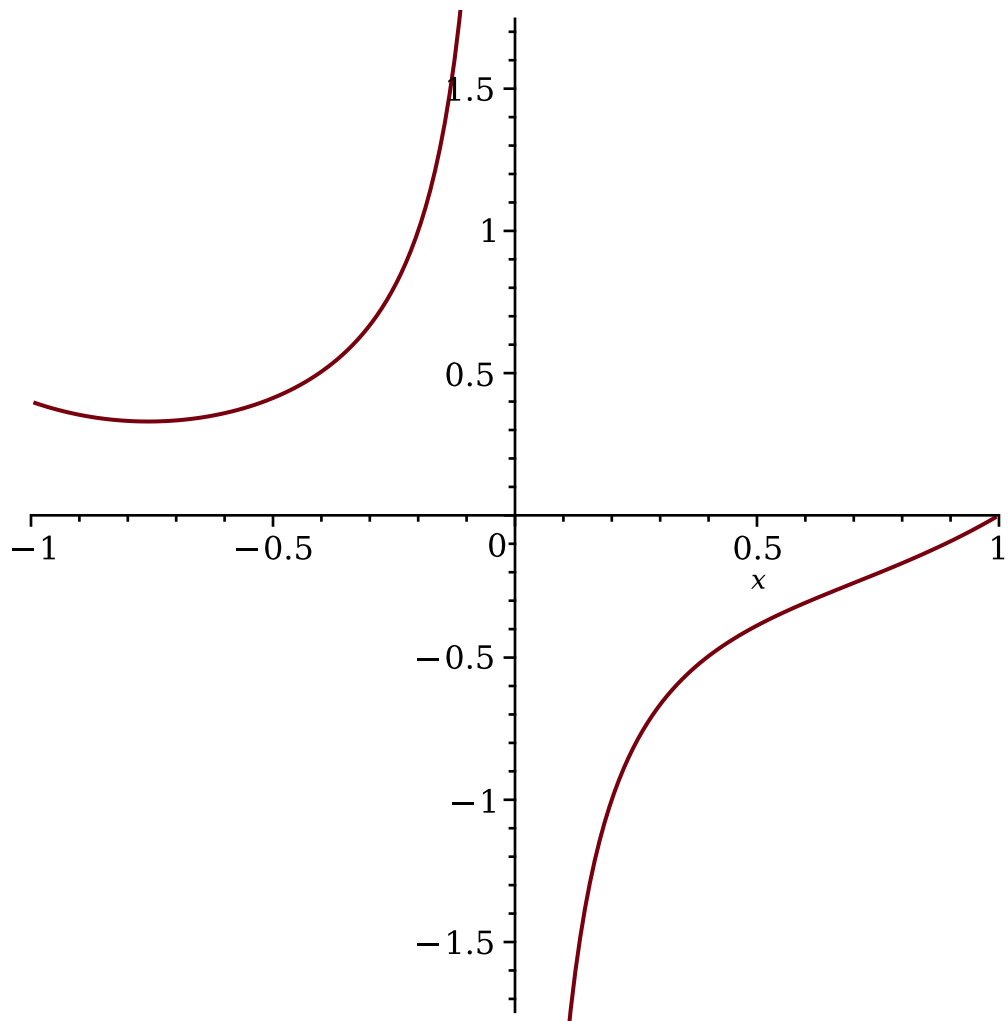
```
> soll2 := dsolve({ecdif2, cond_in2}, y(x));
```

$$soll2 := y(x) = \frac{x^5 - 1}{5x}$$

(29)

```
> y2 := unapply(rhs(soll2), x):
```

```
> plot(y2(x), x = -1 .. 1)
```



```
> soll3 := dsolve({ecdif, cond_in2}, y(x));
```

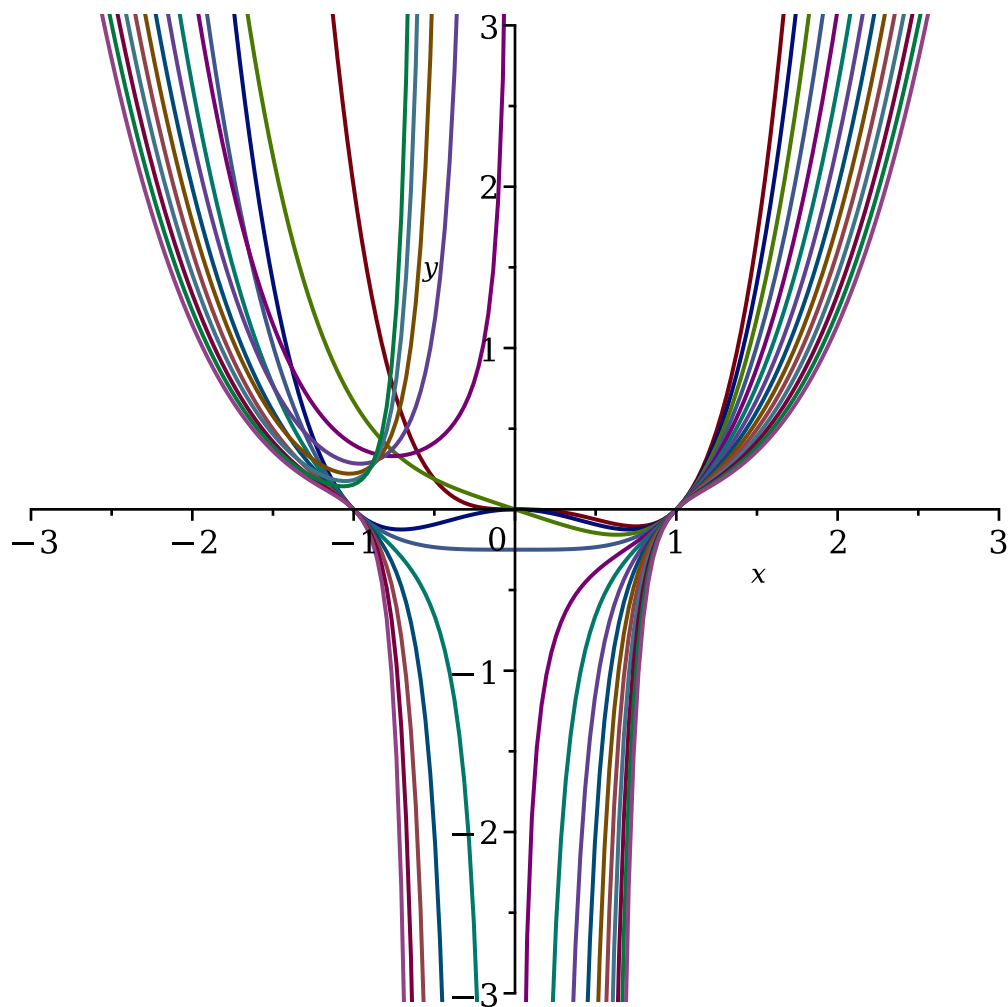
$$soll3 := y(x) = \frac{x^4 - x^{-k}}{4 + k}$$

(30)

```
> y4 := unapply(rhs(soll3), x, k) :
```

```
> list_f2 := y4(x, i) $ i = -3..10 :
```

```
> plot([list_f2], x = -3..3, y = -3..3)
```



```

> restart:
> with(DETools):
> ecdif:=diff(y(x), x, x) - diff(y(x), x) - 2·y(x) = 0:
> cond_in := y(0) = a, D(y)(0) = 2:
> soll := dsolve({ecdif, cond_in}, y(x));
      
$$soll := y(x) = \frac{(-2 + 2a)e^{-x}}{3} + \frac{e^{2x}(a + 2)}{3}$$

(31)
> y1 := unapply(rhs(soll), x, a):
> limit(y1(x, a), x = infinity)
      
$$\text{signum}(a + 2) \infty$$

(32)
> limit(y1(x, -2), x = infinity)
      
$$0$$

(33)
> #a:=-2s

```